

Curriculum Vitae

Personal Information

Name: Shashank Sharma, Ph.D

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Career Objective

I'm interested in synthesising, MOFs and transition-metal-based catalysts that can perform oxygen evolution reaction (OER), hydrogen evolution reaction (HER), complete water splitting, photoelectrochemical water splitting, energy conversion and storage. To contribute as an electrochemistry professional in developing high-performance, durable energy conversion materials and systems that enable clean and efficient renewable energy technologies. I aim to apply my expertise in electrocatalysis, materials design, and electrochemical characterization to support the government's mission of reducing greenhouse gas emissions, advancing energy security, and delivering sustainable, affordable, and reliable energy solutions.

Academic Qualifications

Degree	Institute/University/Board	Year	CGPA/Percentage
Ph.D. (Chemistry)	IISER Bhopal	2020-2025	9.5/10
M.Sc. (Chemistry)	Banaras Hindu University, Varanasi, U.P.	2014-2016	7.69
B.Sc. (Science)	Bundelkhand University, Jhansi, U.P.	2011-2014	75.1%
Intermediate	UP Board, Allahabad	2010	81.0%
High School	UP Board, Allahabad	2008	74.5%

Research Interests

Electrocatalysis for water oxidation, Synthesis of metal organic frameworks (MOFs), Synthesis of Transition metal hydroxides and layered double hydroxides, Two-dimensional (2D) nanomaterials and Nanomesh, Pore engineering and surface reconstruction, Seawater splitting, Water splitting, Photoelectrochemical water splitting, Energy conversion and storage systems, Electrolysers, Fuel cells, and Zn-air batteries.

Ph.D. Research Topic

“Transition Metal-Based Nanomesh for Oxygen Evolution Reaction.”

Summary: Development of ultrathin CoFe-based nanomeshes with tailored porosity and Fe-doping to enhance active site exposure, charge transport, and long-term OER stability.

Supervisor: Prof. Amit Paul, FRSC, Department of Chemistry, IISER Bhopal.

Awards, Travel Grant, and Fellowships

1. Received an outstanding award for poster presentation in “2025 International Conference on Green Electrochemical Technologies (2025-ICGET-Tw), in the Department of Chemical Engineering, NCKU, Tainan, Taiwan (31 October 2025 to 02 November 2025).
2. Received a Travel grant by ANRF-DST, New Delhi, for participating in "The 2025 International Conference on Green Electrochemical Technologies (2025-ICGET-Tw), Taiwan (31 October 2025 to 02 November 2025)". The file number is ITS/2025/004843.
3. MP Young Scientist Award in the 40th M.P. Young Scientist Congress held at Vikram University, Ujjain, and sponsored by M.P. Council of Science and Technology from 27-29 March 2025.
4. ACS Best Poster Award in the 34th CRSI National Symposium in Chemistry held at IISER Bhopal from February 06-08, 2025.
5. RSC Best Poster Award in ChemSci2024: Leaders in the Field Symposium held at IISER Thiruvanthapuram from December 09-11, 2024
6. Best Poster Award in International Conference on Frontiers in Material Engineering 2022 (ICFME-2022) at IIT Indore from December 14-16, 2022.
7. Junior and Senior Research Fellowship (from 01-01-2020 to 31-12-2024) by CSIR, New Delhi.
8. Qualified GATE 2016 exam conducted by IISC Bangalore with AIR-1438 in chemistry.

Technical Skills and Teaching Experience

1. **Material synthesis:** Hydrothermal, Solvothermal, Co-precipitation, Dissolution-regrowth one-pot.
2. **Characterization:** XRD, SEM, TEM, BET, FTIR, AFM, XPS, TGA
3. **Electrochemical techniques:** CV, LSV, EIS, Chronoamperometry, Chronopotentiometry.
4. **Software:** Origin, CHI, EC-Lab, ImageJ, Vesta, Gwyddion, XPSFit, ChemDraw.
5. Scientific writing and an expert in PowerPoint presentations.
6. Teaching Assistant at IISER Bhopal.
7. Guest Lecturer (2017-18) in Nehru Degree College, Lalitpur, Uttar Pradesh, India.

Publications

1. Sharma, S; Paul, A. One-Pot Synthesis of CoFe-Nanomesh for Oxygen Evolution Reaction. *ACS Appl. Nano Mater.* **2024**, 7, 8567–8579.
2. Sharma, S; Paul, A. Evolution of Ultrathin CoFe-Nanomesh for Oxygen Evolution Reaction: From Slit Pores to Ink-Bottle Pores. *Chem. Asian J.* **2025**, 20, e202401156.
3. Sharma, S; and Coworkers; Mo-Induced Pore Expansion in CoFe-Nanomesh for Efficient Oxygen Evolution Reaction. (*Chem. Asian J.* 2026, 21, e70696)

Participation and Talks

1. Participated and presented a Poster in the Conference on Advances in Catalysis for Energy and Environment (CACEE-2022), & CO₂ India Network 1st Annual Meet, held at TIFR Mumbai, from 31st October to 4th November 2022.
2. Participated and presented a poster in the International Conference on Frontiers in Material Engineering 2022 (ICFME-2022) at IIT Indore from December 14-16, 2022.
3. Participated and presented a poster in the International Conference on Nanomaterials for Electro Catalysis Technology (I-CONNECT), at the Indian Institute of Technology, Delhi, from March 20-22, 2023.
4. Participated and presented a poster in ChemSci2024: Leaders in the Field Symposium held at IISER Thiruvanthapuram from December 09-11, 2024.
5. Participated and presented a poster in the 34th CRSI National Symposium in Chemistry held at IISER Bhopal from February 06-08, 2025.
6. Participated and presented an oral in the 40th M.P. Young Scientist Congress held at Vikram University, Ujjain, and sponsored by M.P. Council of Science and Technology from 27-29 March 2025.
7. Participated and presented an oral presentation in the in-house symposium (INTERACTION)-2024 at IISER Bhopal.
8. Participated and presented a poster in the in-house symposium (INTERACTION)-2022, 2023 at IISER Bhopal.
9. Participated in the in-house symposium (INTERACTION)-2020 at IISER Bhopal.
10. Participate and present a poster in the 2025 International Conference on Green Electrochemical Technologies (2025 ICGET-Tw) held on October 31-November 2, 2025, at NCKU, Tainan, Taiwan.

Professional Membership

Student Member, The Royal Society of Chemistry (RSC, 01-01-2025-31-12-2025)

Lifetime Student Member, Chemical Research Society of India (CRSI).

Declaration

I hereby declare that the information given above is genuine to the best of my knowledge.

Date: 31/03/2026


Signature

References

1. Prof. Amit Paul, Professor

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2. Prof. Aasheesh Srivastava, Professor

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3. Prof. Aburba Lal Koner, Professor

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4. Prof. Sangit Kumar, Professor

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